Protocols

IT5100D AY25/26 S2

NUS | Computing
National University of Singapore

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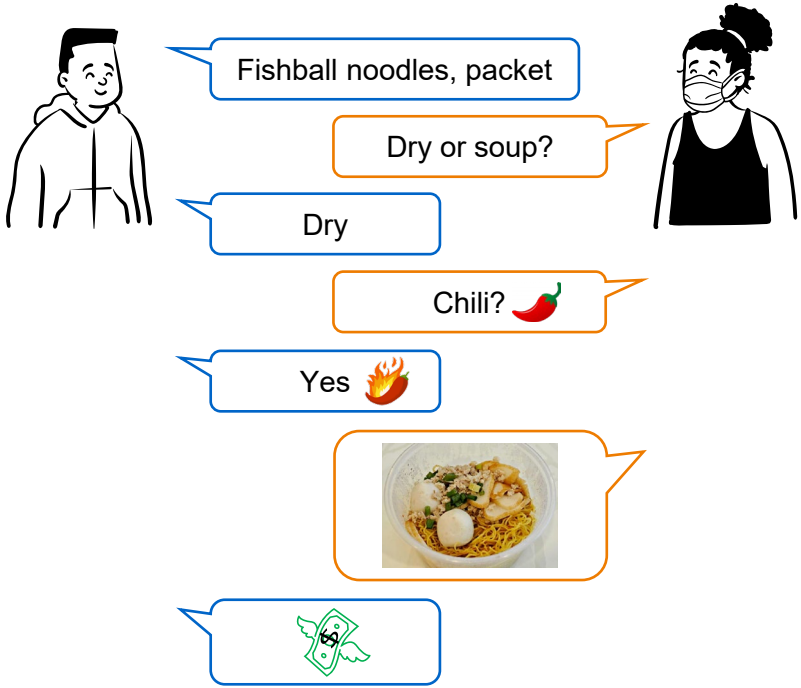
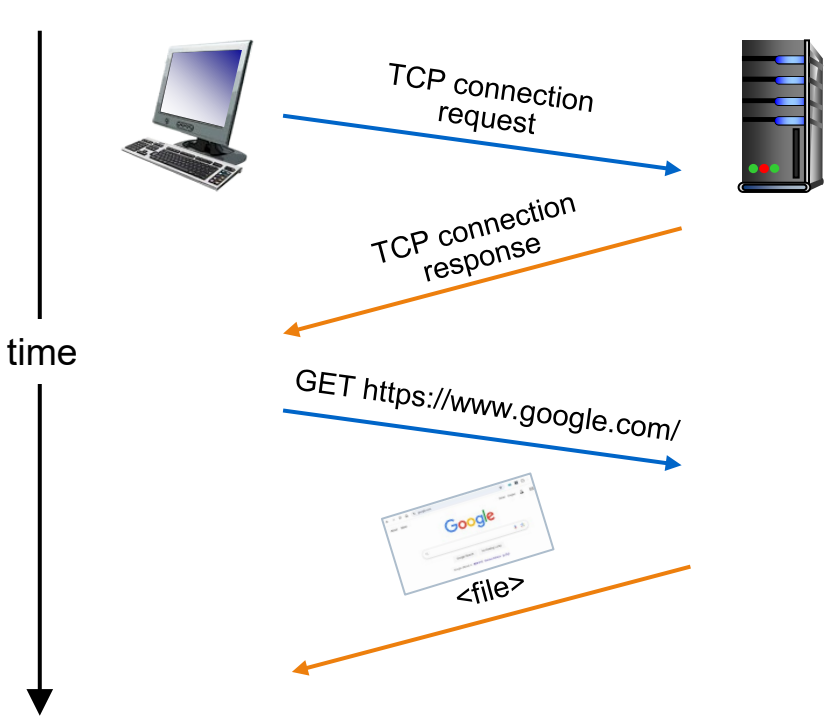
What is a Protocol?

A human protocol:



A computer network protocol:

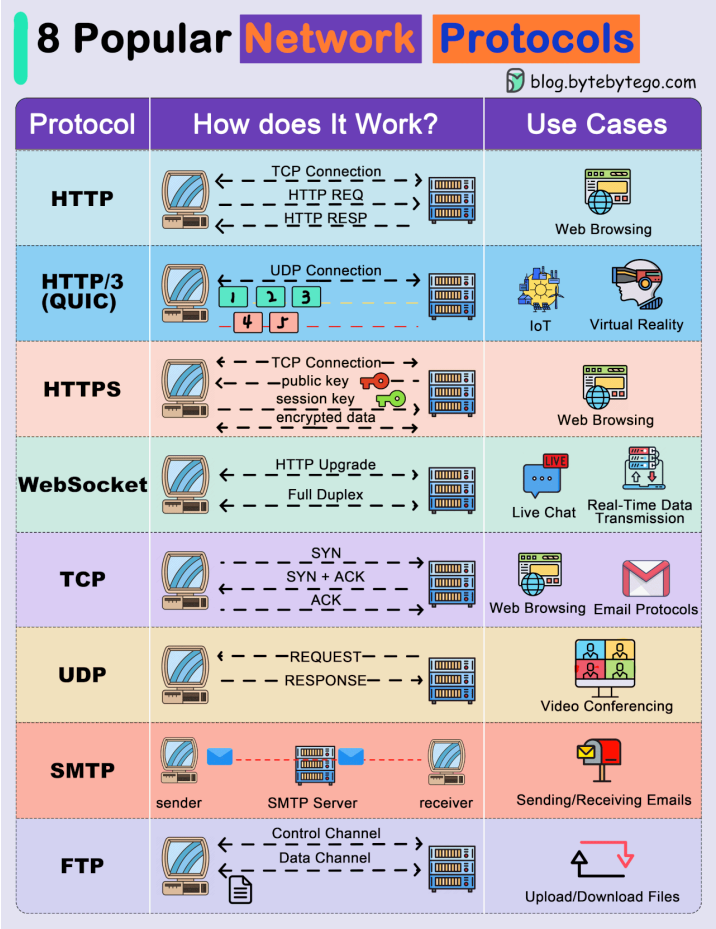


What is a Protocol?

All communication activities on the Internet are governed by **protocols**

Protocols define:

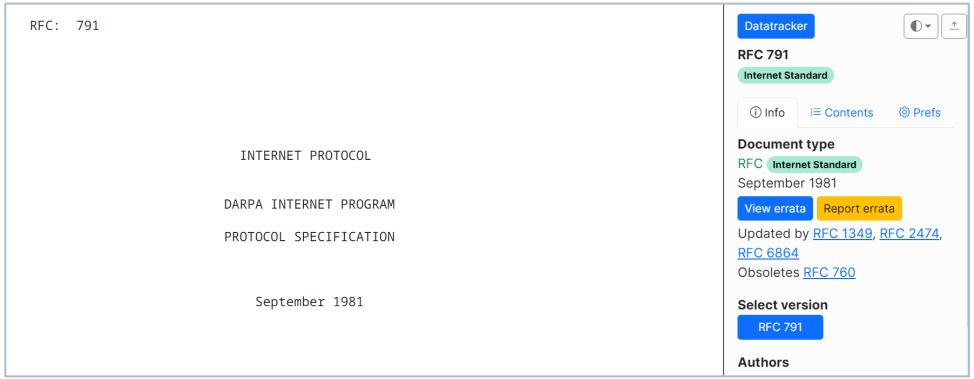
- *Format, order of messages sent and received* among network entities, and
- *Actions taken* on message transmission / receipt



Protocol Standards

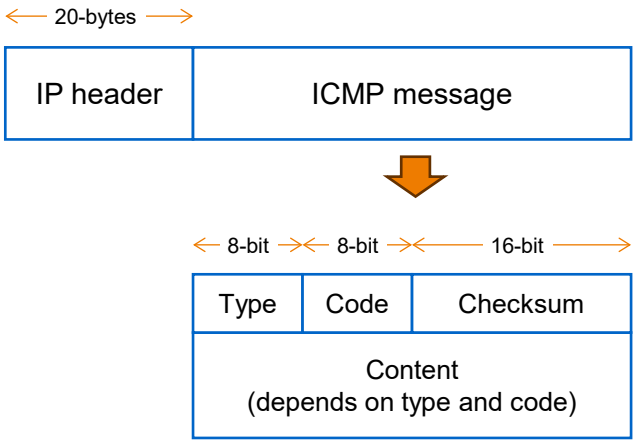
- Standards-setting bodies, most prominently the Internet Engineering Task Force (IETF)
- Internet RFCs (Request for Comments) e.g.,
  - RFC 9293: Transmission Control Protocol (TCP)
  - RFC 791: Internet Protocol (IP)
  - And many more…

- Standards are ultimately specifications, where details may be interpreted differently if there are different implementations.



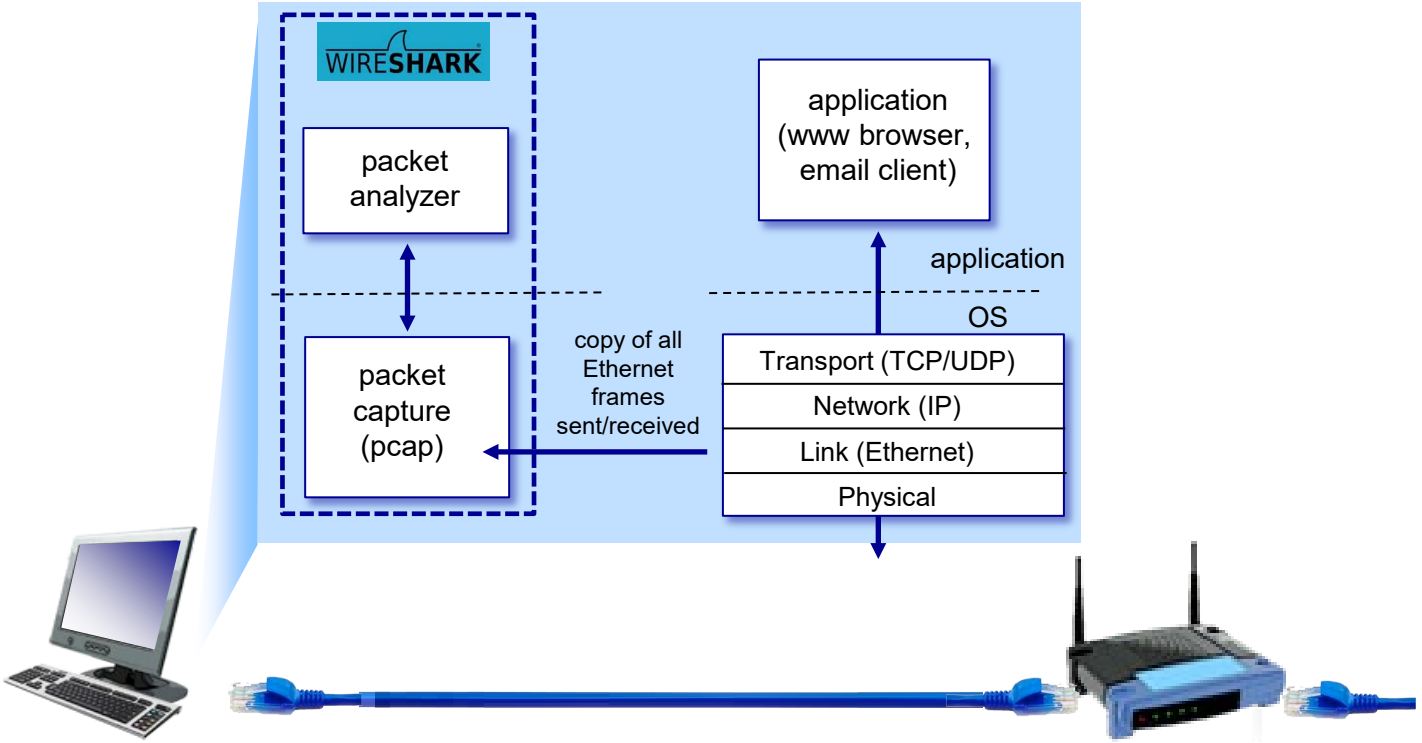
Protocol Example

- ping: a network admin tool to check if a host is reachable on an IP (Internet Protocol) network
  - Manual for Windows, Linux
- It uses Internet Control Message Protocol (ICMP) [RFC 792]



ICMP Type	Code	Description
0	0	echo reply (to ping)
3	0	destination network unreachable
3	1	destination host unreachable
3	2	destination protocol unreachable
3	3	destination port unreachable
3	6	destination network unknown
3	7	destination host unknown
4	0	source quench (congestion control)
8	0	echo request
9	0	router advertisement
10	0	router discovery
11	0	TTL expired
12	0	IP header bad

Let’s See It on Wireshark



Ping on Wireshark

*WiFi

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
151	13.796321	192.168.0.147	45.60.33.225	ICMP	74	Echo (ping) request id=0x0001, seq=74/18944, ttl=128 (reply in 152)
152	14.073125	45.60.33.225	192.168.0.147	ICMP	74	Echo (ping) reply id=0x0001, seq=74/18944, ttl=52 (request in 151)
203	14.812371	192.168.0.147	45.60.33.225	ICMP	74	Echo (ping) request id=0x0001, seq=75/19200, ttl=128 (reply in 210)
210	15.077703	45.60.33.225	192.168.0.147	ICMP	74	Echo (ping) reply id=0x0001, seq=75/19200, ttl=52 (request in 203)
232	15.828244	192.168.0.147	45.60.33.225	ICMP	74	Echo (ping) request id=0x0001, seq=76/19456, ttl=128 (reply in 235)
235	16.094011	45.60.33.225	192.168.0.147	ICMP	74	Echo (ping) reply id=0x0001, seq=76/19456, ttl=52 (request in 232)
250	16.834288	192.168.0.147	45.60.33.225	ICMP	74	Echo (ping) request id=0x0001, seq=77/19712, ttl=128 (reply in 255)
255	17.102611	45.60.33.225	192.168.0.147	ICMP	74	Echo (ping) reply id=0x0001, seq=77/19712, ttl=52 (request in 250)

```
> Frame 151: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{842EEA67-1F87-49F8-871E
> Ethernet II, Src: Intel_80:e6:43 (30:05:05:80:e6:43), Dst: DLinkInterna_f7:2a:c4 (e4:6f:13:f7:2a:c4)
v Internet Protocol Version 4, Src: 192.168.0.147, Dst: 45.60.33.225
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 60
    Identification: 0xda01 (55809)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: ICMP (1)
    Header Checksum: 0x5067 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.147
    Destination Address: 45.60.33.225
v Internet Control Message Protocol
    Type: 8 (Echo (ping) request)
    Code: 0
    Checksum: 0x4d11 [correct]
    [Checksum Status: Good]
    Identifier (BE): 1 (0x0001)
    Identifier (LE): 256 (0x0100)
    Sequence Number (BE): 74 (0x004a)
    Sequence Number (LE): 18944 (0x4a00)
    [Response frame: 152]
  v Data (32 bytes)
      Data: 6162636465666768696a6b6c6d6e6f707172737475767761626364656667686 9
      [Length: 32]
```

```
0000  e4 6f 13 f7 2a c4 30 05  05 80 e6 43 08 00 45 00   ·o··*·0· ···C··E·
0010  00 3c da 01 00 00 80 01  50 67 c0 a8 00 93 2d 3c   ·<------ Pg----<
0020  21 e1 08 00 4d 11 00 01  00 4a 61 62 63 64 65 66   !···M··· ·Jabcdef
0030  67 68 69 6a 6b 6c 6d 6e  6f 70 71 72 73 74 75 76   ghijklmn opqrstuv
0040  77 61 62 63 64 65 66 67  68 69                     wabcdefg hi
```

Command Prompt

```
C:\Users\dcsvivy>ping comp.nus.edu.sg

Pinging comp.nus.edu.sg [45.60.33.225] with 32 bytes of data:
Reply from 45.60.33.225: bytes=32 time=6ms TTL=47
Reply from 45.60.33.225: bytes=32 time=5ms TTL=47
Reply from 45.60.33.225: bytes=32 time=5ms TTL=47
Reply from 45.60.33.225: bytes=32 time=5ms TTL=47

Ping statistics for 45.60.33.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 6ms, Average = 5ms

C:\Users\dcsvivy>
```

Ping on Wireshark

*WiFi

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
151	13.796321	192.168.0.147	45.60.33.225	ICMP	74	Echo (ping) request id=0x0001, seq=74/18944, ttl=128 (reply in 152)
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203	14.812371	192.168.0.147	45.60.33.225	ICMP	74	Echo (ping) request id=0x0001, seq=75/19200, ttl=128 (reply in 210)
210	15.077703	45.60.33.225	192.168.0.147	ICMP	74	Echo (ping) reply id=0x0001, seq=75/19200, ttl=52 (request in 203)
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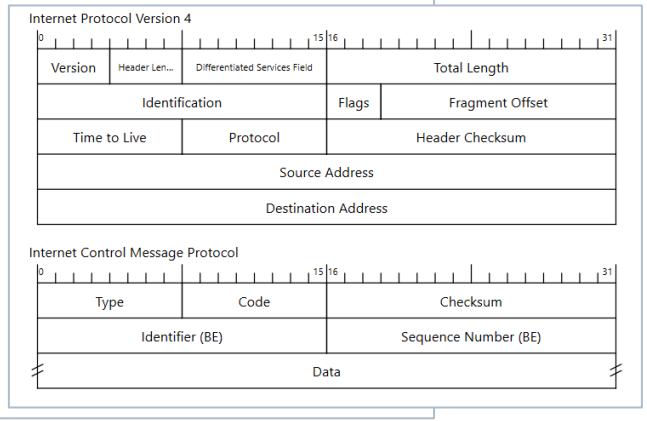
235	16.094011	45.60.33.225	192.168.0.147	ICMP	74	Echo (ping) reply id=0x0001, seq=76/19456, ttl=52 (request in 232)
250	16.834288	192.168.0.147	45.60.33.225	ICMP	74	Echo (ping) request id=0x0001, seq=77/19712, ttl=128 (reply in 255)
255	17.102611	45.60.33.225	192.168.0.147	ICMP	74	Echo (ping) reply id=0x0001, seq=77/19712, ttl=52 (request in 250)

```
> Frame 152: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{842EEA67-1F87-49F8-871E
> Ethernet II, Src: DLinkInterna_f7:2a:c4 (e4:6f:13:f7:2a:c4), Dst: Intel_80:e6:43 (30:05:05:80:e6:43)
v Internet Protocol Version 4, Src: 45.60.33.225, Dst: 192.168.0.147
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x28 (DSCP: AF11, ECN: Not-ECT)
    Total Length: 60
    Identification: 0x2948 (10568)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 52
    Protocol: ICMP (1)
    Header Checksum: 0x4cf9 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 45.60.33.225
    Destination Address: 192.168.0.147
v Internet Control Message Protocol
    Type: 0 (Echo (ping) reply)
    Code: 0
    Checksum: 0x5511 [correct]
    [Checksum Status: Good]
    Identifier (BE): 1 (0x0001)
    Identifier (LE): 256 (0x0100)
    Sequence Number (BE): 74 (0x004a)
    Sequence Number (LE): 18944 (0x4a00)
    [Request frame: 151]
    [Response time: 276.804 ms]
  v Data (32 bytes)
      Data: 6162636465666768696a6b6c6d6e6f707172737475767761626364656667686 9
      [Length: 32]
```

```
0000  30 05 05 80 e6 43 e4 6f  13 f7 2a c4 08 00 45 28   0····C·o ··*···E(
0010  00 3c 29 48 00 00 34 01  4c f9 2d 3c 21 e1 c0 a8   ·<)H··4· L·-<!···
0020  00 93 00 00 55 11 00 01  00 4a 61 62 63 64 65 66   ····U··· ·Jabcdef
0030  67 68 69 6a 6b 6c 6d 6e  6f 70 71 72 73 74 75 76   ghijklmn opqrstuv
0040  77 61 62 63 64 65 66 67  68 69                     wabcdefg hi
```

Wireshark > Preferences > Layout > Packet Diagram



For Something More Interesting…

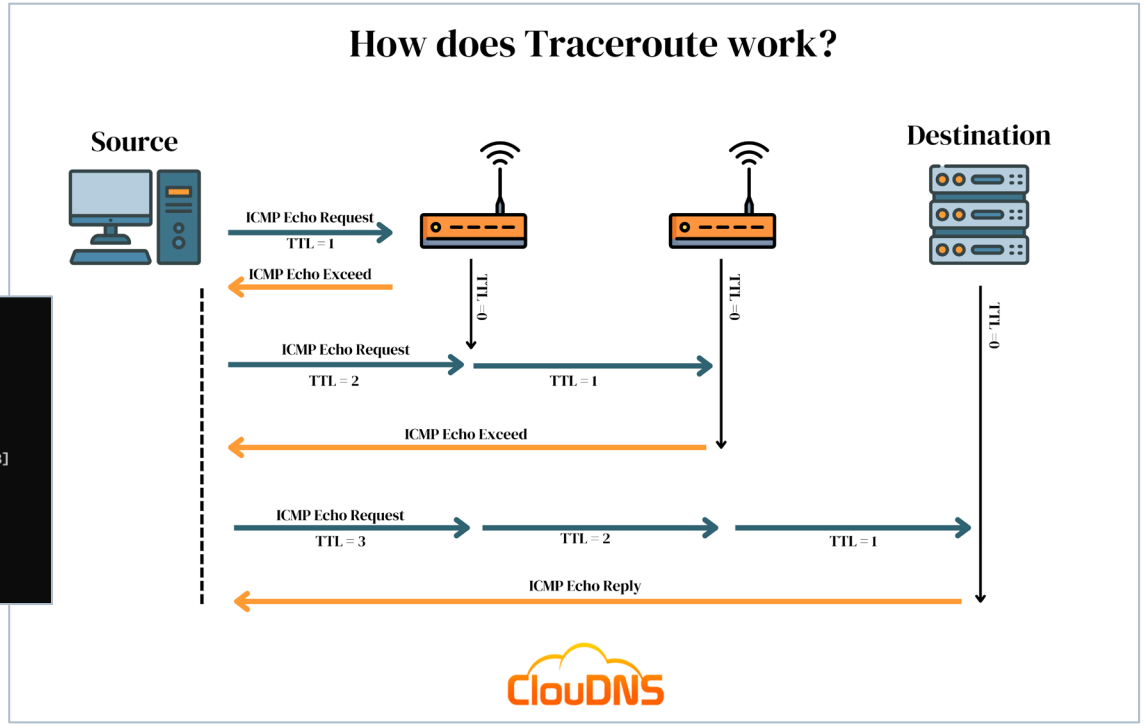
- traceroute: a tool to display the full route (path) to a host, along with round-trip times
  - Manual for Windows (tracert), Linux
- Also uses ICMP – iteratively

```
PS C:\Users\vivy> tracert comp.nus.edu.sg

Tracing route to comp.nus.edu.sg [45.60.33.225]
over a maximum of 30 hops:

  1     4 ms     7 ms     5 ms  1.0.212.218.starhub.net.sg [218.212.0.1]
  2     5 ms     3 ms     7 ms  183.90.44.197
  3     5 ms     3 ms     4 ms  203.118.6.233
  4     6 ms     5 ms     7 ms  203.118.6.149
  5     8 ms     7 ms     4 ms  203.116.3-58.unknown.starhub.net.sg [203.116.3.58]
  6   175 ms   178 ms   174 ms  Frankfurt-MX80.incapsula.com [80.81.193.87]
  7   176 ms   174 ms   173 ms  45.60.33.225

Trace complete.
PS C:\Users\vivy>
```



Traceroute on Wireshark

Capturing from WiFi

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
84	4.163915	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=173/44288, ttl=1 (no response found!)
85	4.167960	218.212.0.1	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
86	4.169469	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=174/44544, ttl=1 (no response found!)
87	4.176516	218.212.0.1	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
88	4.178854	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=175/44800, ttl=1 (no response found!)
89	4.184019	218.212.0.1	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
109	5.200095	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=176/45056, ttl=2 (no response found!)
110	5.204876	183.90.44.197	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
111	5.207988	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=177/45312, ttl=2 (no response found!)
112	5.211646	183.90.44.197	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
113	5.213376	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=178/45568, ttl=2 (no response found!)
114	5.220944	183.90.44.197	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
9321	10.745003	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=179/45824, ttl=3 (no response found!)
9322	10.749868	203.118.6.233	192.168.0.147	ICMP	182	Time-to-live exceeded (Time to live exceeded in transit)
9323	10.751798	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=180/46080, ttl=3 (no response found!)
9324	10.755532	203.118.6.233	192.168.0.147	ICMP	182	Time-to-live exceeded (Time to live exceeded in transit)
9325	10.756739	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=181/46336, ttl=3 (no response found!)
9326	10.760654	203.118.6.233	192.168.0.147	ICMP	182	Time-to-live exceeded (Time to live exceeded in transit)
9388	16.286948	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=182/46592, ttl=4 (no response found!)
9389	16.292829	203.118.6.149	192.168.0.147	ICMP	182	Time-to-live exceeded (Time to live exceeded in transit)
9390	16.294930	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=183/46848, ttl=4 (no response found!)
9391	16.300130	203.118.6.149	192.168.0.147	ICMP	182	Time-to-live exceeded (Time to live exceeded in transit)
9392	16.301457	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=184/47104, ttl=4 (no response found!)
9393	16.308671	203.118.6.149	192.168.0.147	ICMP	182	Time-to-live exceeded (Time to live exceeded in transit)
9409	21.839896	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=185/47360, ttl=5 (no response found!)
9410	21.847773	203.116.3.58	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
9411	21.850417	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=186/47616, ttl=5 (no response found!)
9412	21.858123	203.116.3.58	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
9413	21.859353	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=187/47872, ttl=5 (no response found!)
9414	21.863862	203.116.3.58	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
9439	22.878613	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=188/48128, ttl=6 (no response found!)
9440	23.053998	80.81.193.87	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
9441	23.056794	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=189/48384, ttl=6 (no response found!)
9442	23.235073	80.81.193.87	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
9443	23.238033	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=190/48640, ttl=6 (no response found!)
9444	23.412715	80.81.193.87	192.168.0.147	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
9445	24.259537	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=191/48896, ttl=7 (reply in 9448)
9448	24.435846	45.60.33.225	192.168.0.147	ICMP	106	Echo (ping) reply id=0x0001, seq=191/48896, ttl=54 (request in 9445)
9449	24.438543	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=192/49152, ttl=7 (reply in 9450)
9450	24.613314	45.60.33.225	192.168.0.147	ICMP	106	Echo (ping) reply id=0x0001, seq=192/49152, ttl=54 (request in 9449)
9451	24.616467	192.168.0.147	45.60.33.225	ICMP	106	Echo (ping) request id=0x0001, seq=193/49408, ttl=7 (reply in 9452)
9452	24.790103	45.60.33.225	192.168.0.147	ICMP	106	Echo (ping) reply id=0x0001, seq=193/49408, ttl=54 (request in 9451)
12260	197.314252	192.168.0.147	40.99.26.178	ICMP	44	Echo (ping) request id=0x0001, seq=194/49664, ttl=30 (reply in 12273)
12273	197.407679	40.99.26.178	192.168.0.147	ICMP	44	Echo (ping) reply id=0x0001, seq=194/49664, ttl=238 (request in 12260)

> Frame 84: 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on inter
> Ethernet II, Src: Intel_80:e6:43 (30:05:05:80:e6:43), Dst: DLinkInterna_f7:2a:

```
0000  e4 6f 13 f7 2a c4 30 05  05 80 e6 43 08 00 45 00   ·o··*·0· ···C··E·
0010  00 5c 78 47 00 00 01 01  31 02 c0 a8 00 93 2d 3c   ·\xG···· 1·····-<
0020  21 e1 08 00 f7 51 00 01  00 ad 00 00 00 00 00 00   !····Q·· ********
```

WiFi: <live capture in progress> | Packets: 21262 · Displayed: 44 (0.2%) | Profile: Default

```
PS C:\Users\vivy> tracert comp.nus.edu.sg

Tracing route to comp.nus.edu.sg [45.60.33.225]
over a maximum of 30 hops:

  1     4 ms     7 ms     5 ms  1.0.212.218.starhub.net.sg [218.212.0.1]
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  4     6 ms     5 ms     7 ms  203.118.6.149
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Trace complete.
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